H-Bit Protocol

Universal Cryptographic Steganography

for Content Authenticity in the Age of AI

Version 1.1.0 -- June 2026

github.com/Ping-iop/H-Bit

ABSTRACT

H-Bit is a universal protocol for establishing an inalienable link between intellectual authorship and any digital file. It embeds Ed25519 cryptographic signatures directly into pixel, frequency, or stream data at amplitudes below the Just Noticeable Difference (JND) threshold -- invisible to humans, readable by machines. Unlike traditional binary verification (authentic/tampered), H-Bit introduces Spectrum Verification: a confidence spectrum (0-100%) that quantifies authenticity evidence even from partial fragments. On a 512x512 image, H-Bit achieves PSNR >55 dB, maintains AUTHENTIC verdict with only 3% of the image data, and correctly identifies AI-generated vs human-created content.

1. Introduction

The rapid advancement of generative AI has created an unprecedented crisis in content authenticity. Deepfakes, AI-generated images, synthetic audio, and LLM-produced text can no longer be reliably distinguished from human-created content. Traditional approaches -- cryptographic hashes, metadata signatures, blockchain notarization -- all require the ENTIRE original file. A single bit flip, crop, recompression, or metadata stripping renders these methods useless.

H-Bit addresses this gap with steganographic embedding of cryptographic signatures directly into the content itself, at signal amplitudes below human perceptual thresholds. This creates an inalienable bond between content and authorship that survives partial corruption and provides a confidence SPECTRUM rather than a binary verdict.

2. System Architecture

H-Bit employs a multi-layer architecture:

```
[Raw File] -> Format Manager -> Crypto Core -> Payload Builder
                                  |
                        Resilience (ECC + Tiling)
                                  |
                  Embedder (LSB / DCT / Stream)
                                  |
                      [Signed File]
                                  |
                 Spectrum Verifier (0-100%)
```

3. Spectrum Verification (Novel Contribution)

The key innovation of H-Bit v1.1.0 is Spectrum Verification: a non-binary confidence model (0-100%). Traditional hash verification is brittle -- a single modified pixel produces TAMPERED. H-Bit examines EACH embedded tile independently and computes a weighted confidence score from five metrics.

Confidence formula: $C = 0.30*R + 0.25*S + 0.20*E + 0.15*Q + 0.10*P$

R = Recovery Rate (valid/total tiles), S = Author Consensus (agreement among tiles), E = ECC Health (fewer corrections = better), Q = Sync Quality (Barker correlation), P = Payload Completeness (structure intact vs partial).

3.1 Verdict Levels

Confidence	Verdict	Interpretation
>= 95%	AUTHENTIC	Complete verification, all tiles agree
>= 75%	LIKELY_AUTHENTIC	High confidence, minor inconsistencies
>= 50%	POSSIBLY_AUTHENTIC	Moderate confidence, some evidence
>= 25%	UNCERTAIN	Insufficient evidence for conclusion
< 25%	LIKELY_TAMPERED	Evidence suggests manipulation
0%	NO_EVIDENCE	No H-Bit signature detected

4. Experimental Results

4.1 Quality Metrics (512x512 PNG)

Metric	Value	Notes
PSNR	55.9 dB	>40 dB = imperceptible (excellent)
SSIM	1.0000	Structurally identical to original
Bits Embedded	798	107-byte core + 78-bit sync wrapper
Capacity Used	99.8%	Near-full LSB channel utilization

4.2 Speed Performance

Operation	Time	Notes
Encode	114.7 ms	Keygen + payload + LSB embedding
Decode	228.6 ms	Bit extraction + deserialization
Spectrum	708.3 ms	Full tile-by-tile analysis
File Overhead	0 bytes	PNG lossless: no size increase

4.3 Crop Robustness (height-based, same width)

Crop %	Rows	Verdict	Confidence	Tiles
100%	512	AUTHENTIC	98%	328/328
75%	384	AUTHENTIC	98%	246/246
50%	256	AUTHENTIC	98%	164/164
25%	128	AUTHENTIC	98%	82/82
12%	61	AUTHENTIC	98%	39/39
6%	30	AUTHENTIC	98%	19/19
3%	15	AUTHENTIC	98%	9/9

Key finding: H-Bit maintains AUTHENTIC verdict with 98% confidence even when only 3% of the original image data is available (15 rows out of 512). Each embedded tile carries the complete payload, so any surviving tile provides full verification.

4.4 Deepfake Detection

Scenario	Verdict	Confidence	Origin
Real Photo (100%)	AUTHENTIC	98%	HUMAN
AI Image (100%)	AUTHENTIC	100%	AI_GENERATED
Real Photo (25% crop)	AUTHENTIC	98%	HUMAN
AI Image (25% crop)	AUTHENTIC	100%	AI_GENERATED
Unsigned Image	NO_EVIDENCE	0%	N/A

5. Cryptographic Foundation

Ed25519 (RFC 8032) provides 128-bit security with 32-byte public keys and 64-byte signatures. HKDF (RFC 5869) with SHA-256 handles key derivation. AES-256-GCM with PBKDF2 (100,000 iterations) encrypts payloads.

Payload structure (107 bytes core):

```
[Version:1B][Flags:1B][OriginType:1B][AuthorHash:32B]
[ContentHash:32B][Timestamp:8B][AIModelID:32B]
+ optional [Signature:64B][ECC:variable]
```

Origin types: HUMAN (0x00), AI_GENERATED (0x01), AI_ASSISTED (0x02), UNKNOWN (0xFF). Optional zlib compression reduces embedded size when payload structure permits.

6. Resilience Engineering

Barker-13 Synchronization: 39-bit composite sync markers with optimal autocorrelation. Tolerates ~15% bit errors while maintaining detectable peaks.

Reed-Solomon ECC: Four presets (light=5, standard=10, heavy=16, forensic=25 symbol error correction). Parity bytes adaptively selected based on expected degradation.

Cyclic Tiling: Payload repeated across entire medium with interleaving to distribute damage. Each tile = complete payload copy. Any surviving tile = full verification.

DCT Watermarking (JPEG mode): Quantization Index Modulation in mid-frequency coefficients, constrained by Watson JND perceptual model for invisibility.

7. Universal Format Support

Format	Strategy	Handler
PNG, BMP, TIFF, WebP	LSB (Blue channel)	ImageHandler
JPEG	DCT (Green)	ImageHandler
WAV, FLAC, AIFF	LSB in PCM	AudioHandler
MP4, AVI, MOV	LSB in keyframes	VideoHandler
PDF	Hidden stream object	PDFHandler
DOCX, XLSX, PPTX	Custom XML part	OfficeHandler
Any other	Append stream + CRC32	GenericHandler

8. Applications

Content Provenance: Platforms display verified badges (HUMAN/AI) surviving crops.
Journalism: Sign at capture time, verify from screenshots or partial crops.
Social Media: Automated AI-content detection with cryptographic proof.
Legal/Evidence: Chain-of-custody from any file fragment to original author.
Hardware: Camera ISP enclave, USB HSM signer, flash FTL driver for zero-touch signing.

9. Conclusion

H-Bit v1.1.0 provides production-ready content authenticity for the AI era. Key differentiators: (1) Non-binary spectrum verification, (2) Universal format support through pluggable handlers, (3) Resilience to partial data loss through tiling and ECC, (4) Cryptographic binding of origin type (HUMAN/AI) directly into content. Open-source (Apache 2.0), 185+ tests, ready for integration into platforms, cameras, and content management systems.

Future work: Native C/Rust SDK for embedded systems, browser extension for real-time web verification, C2PA manifest integration, hardware partnership program, and formal standardization through W3C/ISO.

10. References

[1] H-Bit Repository: github.com/Ping-iop/H-Bit
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